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All Polymer Diffusion Regimes Covered by Combining Field-Cycling and Field-Gradient ^{1}H NMR

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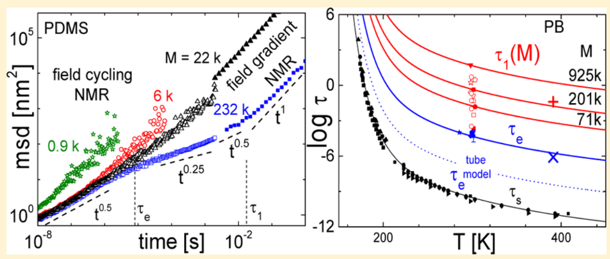
ABSTRACT: Field-cycling and field-gradient ^{1}H NMR experiments were combined to reveal the segmental mean-square displacement as a function of time for polydimethylsiloxane (PDMS) and polybutadiene (PB). Together, more than 10 decades in time are covered, and all four power-law regimes of the tube-reptation (TR) model are identified with exponents rather close to the predicted ones. Characteristic polymer properties like the tube diameter a_0, the Kuhn length b, the mean-square end-to-end distance $\langle R_0^2 \rangle$, the segmental correlation time $\tau_s(T)$, the entanglement time $\tau_e(T)$, and the disengagement time $\tau_d(T)$ are estimated from the measurements and compared to results from literature. Concerning $\tau_d(T)$, fair agreement is found. In the case of τ_e, agreement with rheological data is achieved when the time constant is extracted from the minimum in the shear modulus $G''(\omega)$. Concerning the TR predictions the molar mass (M) dependence of τ_d is essentially reproduced. Yet, calculating τ_e from τ_d for PDMS yields agreement with experimental data while for PB it gets by 2 orders of magnitude too short. In no case τ_e is correctly reproduced from $\tau_s(T)$. Segmental and shortest Rouse times appear to coincide for PB, while in the case of PDMS the latter turns out to be longer by 1 decade.



1. INTRODUCTION

The tube-reptation (TR) model[1] is a widely accepted approach to describe the microscopic dynamics of entangled polymers melts. The model predicts four different power-law regimes (I–IV) for the time dependence of the segmental mean-square displacement (msd) $\langle r^2(t) \rangle \propto t^\alpha$ (cf. inset of Figure 5b). Molecular dynamics simulations[2,3] and neutron scattering (NS) experiments[4] essentially confirmed parts of the model by identifying the predicted crossover from $\langle r^2(t) \rangle \propto t^{0.5}$ to $\langle r^2(t) \rangle \propto t^{0.25}$ for the transition from free Rouse (regime I) to constrained Rouse dynamics (regime II). On long time scales the crossover from $\langle r^2(t) \rangle \propto t^{0.5}$, characteristic for reptation (regime III), to $\langle r^2(t) \rangle \propto t^1$, for the terminal regime of free diffusion (regime IV), was observed by field gradient (FG) NMR.[5−8] However, the agreement was only qualitative since numerical discrepancies with respect to the TR model occurred. Another regime (regime 0) reflects local motions connected with fluctuations for times on the order of the segmental correlation time τ_s determined by structural relaxation of the glass transition dynamics. Given the large range of relaxation regimes, characterization by essentially a single experiment is still missing.

Recently, field-cycling (FC) ^{1}H NMR relaxometry[9,10] confirmed the crossover between the power-law behavior of regime I and II and has thus become a complementary method to FG NMR and a competing one to NS. The method, which measures the frequency dependence (dispersion) of the spin–lattice relaxation rate $R_1 = 1/T_1$, gained momentum with the availability of advanced FC relaxometers.[9−16] Commercial machines typically cover a frequency range of 10 kHz–20 MHz. Using a home-built relaxometer, where we applied active earth and stray field compensation, we are able to achieve a low-frequency limit of about 100 Hz (in ^{1}H Larmor frequency units).[10,14−16] In the case of ^{1}H NMR the relaxation originates from fluctuations of the magnetic dipole–dipole interaction. Here one has to distinguish contributions from spins belonging to the same polymer molecule constituting the intramolecular contribution, and spins from different molecules, representing the intermolecular contribution.[17] As a consequence, the total relaxation rate is given by the sum of both contributions, i.e., $R_1 = R_1^{\text{intra}} + R_1^{\text{inter}}$. In polymer melts $R_1^{\text{inter}}(\omega)$ is dominated by translational dynamics and hence reflects the relative msd of segments from different macromolecules.[17] Assuming sub-diffusive power-law governed translational dynamics for times $t < \tau_1$, where τ_1 is the terminal relaxation time,[1] it was demonstrated that $\langle r^2(t) \rangle$ can be estimated from the intermolecular relaxation rate $R_1^{\text{inter}}(\omega)$ by a simple analytical expression (cf. Theoretical Background).[9,10,18] Note that in the framework of the TR theory the terminal relaxation time τ_1 is equivalent to the tube disengagement time τ_d. The intermolecular relaxation rate was singled out from the total relaxation by applying isotope dilution experiments for which

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protonated chains are successively diluted in a perdeuterated polymer matrix.[9,10,19−21]

A more general relationship connecting $\langle r^2(t)\rangle$ directly with the Fourier transform of $R_1^{\text{inter}}(\omega)$ was derived[9,18] which does not rely on any assumptions (cf. Theoretical Background). It is the purpose of the present contribution to test this approach and to compare it with results measured by FG ^{1}H NMR which provides absolute values of the segmental msd in the regime III and IV. The method is a well-established technique to access translational diffusion; $\langle r^2(t)\rangle$ is usually probed in the hydrodynamic limit (regime IV in the case of polymers).[6] Yet, in order to access also shorter times $t < \tau_1$, where the diffusion coefficient becomes time dependent, special experimental efforts have to be undertaken. In particular, the so-called dipolar correlation effect due to reorientational motion has to be accounted for.[6,22] We will demonstrate, for the first time, that all four regimes of polymer diffusion are covered by combining data of two NMR methods for polybutadiene (PB) and polydimethylsiloxane (PDMS). Moreover, a comparison with rheological data is carried out and discussed within the frame of the TR model.

2. THEORETICAL BACKGROUND

2.1. Mean-Square Displacement Derived from ^{1}H FC NMR. Here we provide a derivation of the relationship between the intermolecular spin−lattice relaxation time $T_1(\omega)$ accessible by FC ^{1}H NMR and the segmental msd. Proton relaxation is determined by fluctuations of the magnetic dipole−dipole interaction, specifically, by the following correlation function:

$$A_p(t) = \frac{1}{N_s}\sum_{k\neq m}\left\langle \frac{Y_{2p}(\vec{e}_{km}(t))}{r_{km}{}^3(t)}\frac{Y_{2p}^*(\vec{e}_{km}(0))}{r_{km}{}^3(0)}\right\rangle \tag{1}$$

$Y_{lp}(\vec{e}_{km})$ is component p of the spherical function of rank l, N_s is the total number of spins in the system, r_{km} is the internuclear distance, and $\vec{e}_{km}(t) = \vec{r}_{km}(t)/r_{km}(t)$ is the corresponding unit vector. In isotropic systems the correlation function becomes independent of p, yielding $A_0(t) = A_1(t) = A_2(t) \equiv A(t)$.[11,17] The correlation function includes intra- and intermolecular contributions; i.e., it comprises correlations between spins belonging to the same macromolecule and to spins from different macromolecules, respectively. We note that we use the nomenclature $A(t)$ in the present context denoting the reorientational correlation function while we usually chose $C(t)$.[10,13,19]

The relaxation rate $R_1 = T_1^{-1}$ is a linear combination of Fourier transforms of $A(t)$ evaluated at the frequencies ω and 2ω:[17]

$$\frac{1}{T_1(\omega)} = \left(\frac{\mu_0}{4\pi}\right)^2\frac{6\pi}{5}\gamma_H{}^4\hbar^2\int_0^\infty dt\{A(t)\cos(\omega t) + 4A(t)\cos(2\omega t)\} \tag{2}$$

Note that this expression is only correct as long as the so-called Redfield condition is satisfied, i.e.,$T_1(\omega) \gg 2\pi\omega^{-1}$. At very low frequencies ($\nu = \omega/2\pi < 400$ Hz) this limit is approached for high M in the present case. A violation of the Redfield condition would go along with a pseudo-Gaussian magnetization decay, which is actually not (yet) observed. Instead, all magnetization curves encountered in the course of this work are monoexponential. Furthermore, a common spin temperature has to be established by fast flip-flop processes (spin diffusion) which also applies well within the frequency window ν = 200 Hz−30 MHz covered by our experiments.

Equation 2 can be rewritten

$$\frac{1}{T_1(\omega)} = \left(\frac{\mu_0}{4\pi}\right)^2\frac{6\pi}{5}\gamma_H{}^4\hbar^2\int_0^\infty dt\cos(\omega t)\{A(t) + 2A(t/2)\} \tag{3}$$

The Fourier transform in eq 3 connects the total dipolar correlation function, including intra- and intermolecular contributions, with the dispersion of the relaxation rate:

$$\hat{A}(t) = A(t) + 2A(t/2) = \left(\frac{4\pi}{\mu_0}\right)^2\frac{5}{3\pi^2}\frac{1}{\gamma_H{}^4\hbar^2}\int_0^\infty d\omega\,\frac{\cos(\omega t)}{T_1(\omega)} \tag{4}$$

The total dipolar correlation function can be decomposed into a sum of inter- and intramolecular parts:[11,17]

$$\hat{A}(t) = \hat{A}^{\text{inter}}(t) + \hat{A}^{\text{intra}}(t) \tag{5}$$

The intermolecular part reads

$$\begin{aligned}\hat{A}^{\text{inter}}(t) &= A^{\text{inter}}(t) + 2A^{\text{inter}}(t/2)\\ &= \left(\frac{4\pi}{\mu_0}\right)^2\frac{5}{3\pi^2}\frac{1}{\gamma_H{}^4\hbar^2}\int_0^\infty d\omega\,\frac{\cos(\omega t)}{T_1^{\text{inter}}(\omega)}\end{aligned} \tag{6}$$

For times $t \gg \tau_s$ $A^{\text{inter}}(t)$ assumes a universal form valid for all models of polymer dynamics, explicitly

$$A^{\text{inter}}(t) = \frac{4\pi}{9}n_s\tilde{W}(0;\,t) \tag{7}$$

where n_s is the proton spin density and $\tilde{W}(\vec{r}',\vec{r};t) = \tilde{W}(\vec{r}'-\vec{r};t)$ the propagator of relative displacements of two spins from different macromolecules, i.e., the probability density for two spins separated by a vector $\vec{r}$ initially and by a vector $\vec{r}'$at later time t. For times $t \gg \tau_s$ the $A^{\text{inter}}(t)$ is proportional to the probability density to recover the initial spatial separation at time t, i.e., the vector $\vec{r} = \vec{r}'$.[18] Next, the propagator in eq 7 can be approximated by a Gaussian distribution:[23]

$$\tilde{W}(r;\,t) = ((2\pi/3)\langle\tilde{r}^2(t)\rangle)^{-3/2}\exp\left\{-\frac{3}{2}\frac{r^2}{\langle\tilde{r}^2(t)\rangle}\right\} \tag{8}$$

Here $\langle\tilde{r}^2(t)\rangle = \langle((\vec{r}_i(t) - \vec{r}_i(0)) - (\vec{r}_j(t) - \vec{r}_j(0)))^2\rangle$ is the relative msd of two arbitrary spins (i, j) located on different macromolecules. Combining eqs 7 and 8 and setting $\vec{r} = 0$ yields

$$A^{\text{inter}}(t) = \sqrt{\frac{2}{3\pi}}\frac{n_s}{(\langle\tilde{r}^2(t)\rangle^{1/2})^3} \tag{9}$$

Thus, for times $t \gg \tau_s$ one is left with two parameters governing the intermolecular correlation function: the spin density n_s and the relative msd $\langle\tilde{r}^2(t)\rangle$. In this case $\langle\tilde{r}^2(t)\rangle^{1/2} \gg \sigma^*$ holds with $\sigma^* \propto (b^2\rho_s)^{-1}$ being a microscopic quantity characterizing the spatial separation of nearest segments; b is the Kuhn length, and ρ_s is the concentration of Kuhn segments. Otherwise, if the time scale approaches the segmental correlation time $t \cong \tau_s$, the relative msd becomes of the order of the segmental packaging length, i.e., $\langle\tilde{r}^2(t)\rangle^{1/2} \cong \sigma^*$, and eq 9 needs to be corrected:

$$A^{\text{inter}}(t) = \sqrt{\frac{2}{3\pi}}\frac{n_s}{(\sigma^* + \langle\tilde{r}^2(t)\rangle^{1/2})^3} \tag{10}$$

We emphasize that the Gaussian approximation (8) does not necessarily assume that diffusion has Brownian character, i.e.,

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$\langle \tilde{r}^2(t)\rangle \propto t$, but also applies for subdiffusive motion. By substituting eq 9 into eq 6, a connection between the relative msd and the relaxation rate $(1/(T_1^{\text{inter}}(\omega)))$ can be established:

$$\frac{1}{\langle \tilde{r}^2(t)\rangle^{3/2}} + \frac{2}{\langle \tilde{r}^2(t/2)\rangle^{3/2}} = \frac{5}{4n_s}\left(\frac{4\pi}{\mu_0}\right)^2 \sqrt{\frac{8}{3\pi^3}} \frac{1}{\gamma_H^4 \hbar^2} \int_0^\infty d\omega \frac{\cos(\omega t)}{T_1^{\text{inter}}(\omega)} \tag{11}$$

In polymer melts translational motion is subdiffusive on time scales $\tau_s \le t \le \tau_1$ and the msd scales as $\langle r^2(t)\rangle \propto t^\alpha$, where the exponent $\alpha \in [0.25, 1]$, depending on the regime of the underlying polymer dynamics ($\alpha = 1$ reflects isotropic diffusion). Exploiting this power-law character of the msd, the left-hand side of eq 11 can be rewritten

$$\frac{1}{\langle \tilde{r}^2(t)\rangle^{3/2}} + \frac{2}{\langle \tilde{r}^2(t/2)\rangle^{3/2}} = \frac{1 + 2^{1+3\alpha/2}}{\langle \tilde{r}^2(t)\rangle^{3/2}} \tag{12}$$

Note that the α-dependent factor (which is denoted as δ from now on) in eq 12 changes rather slowly for $\alpha \in [0.25, 1]$, thus varying within $3.6 \le \delta := 1 + 2^{1+(3\alpha/2)} \le 6.6$. Finally, to convert the actually measured relative msd $\langle \tilde{r}^2(t)\rangle$ into the absolute displacement $\langle r^2(t)\rangle$, one assumes that motion of chains is essentially independent of each other. Then the relative segmental msd can be estimated as twice as large as the absolute msd yielding the final result

$$\langle r^2(t)\rangle = \frac{1}{2}\langle \tilde{r}^2(t)\rangle = \frac{1}{2}\left[\frac{5}{4}\left(\frac{4\pi}{\mu_0}\right)^2 \sqrt{\frac{8}{3\pi^3}} \frac{1}{\gamma_H{}^4 \hbar^2 n_s} \frac{1}{\delta} \int_0^\infty d\omega \frac{\cos(\omega t)}{T_1^{\text{inter}}(\omega)}\right]^{-2/3} \tag{13}$$

This expression will be applied in the present study to derive the msd.

For a constant $\alpha < (2/3)$ eq 13 is equivalent to

$$\langle r^2(t = \omega^{-1})\rangle = \frac{1}{2}\langle \tilde{r}^2(t = \omega^{-1})\rangle = \frac{1}{2}\left[\frac{5}{12}\left(\frac{4\pi}{\mu_0}\right)^2 \sqrt{\frac{6}{\pi}} \frac{1}{\gamma_H{}^4 \hbar^2 n_s} \frac{1}{f(a)} \frac{\omega}{T_1^{\text{inter}}(\omega)}\right]^{-2/3} \tag{14}$$

which was originally derived in ref 9 with

$$f(\alpha) = \frac{\pi\delta}{2\cos\left(\frac{3\pi\alpha}{4}\right)\Gamma\left(\frac{3\alpha}{2}\right)} \tag{15}$$

Assuming $\langle r^2(t)\rangle \propto t^\alpha$, eq 14 directly leads to

$$\frac{1}{T_1^{\text{inter}}(\omega)} \propto \frac{1}{\omega^{3/2\alpha - 1}} \tag{16}$$

For calculating the msd as a function of time, $R_1^{\text{inter}}(\omega)$ is required over a wide range of frequencies. The long-time diffusion coefficient D, however, can be obtained from the low-frequency limit of the total relaxation dispersion $R_1(\omega)$.[24,25] This was recently demonstrated for simple liquids[26] as well as for polymers.[27] The low-frequency dispersion of the rate $R_1(\omega)$ is always dominated by the intermolecular relaxation contribution mediated by translational motion, while at higher frequencies both inter- and intramolecular contributions are relevant. In a liquid, self-diffusion at long times is Fickian, yielding a power-law $A^{\text{inter}}(t) \propto t^{-3/2}$. Therefore, the total relaxation rates (including intra- and intermolecular contributions) can be expanded at low frequencies, providing a universal dispersion law:[24−27]

$$R_1(\omega) = R_1^{\text{intra}}(\omega) + R_1^{\text{inter}}(\omega) = R_1(0) - \frac{B}{D^{3/2}}\sqrt{\omega} \tag{17}$$

with

$$B = \frac{\pi}{30}(1 + 4\sqrt{2})\left(\frac{\mu_0}{4\pi}\hbar\gamma_H{}^2\right)^2 n_s$$

The intramolecular (reorientational) contribution is included in $R_1(0)$ as the rotational contribution is frequency independent at low frequencies ($\omega\tau_{\text{rot}} \ll 1$). The corresponding spectral density is constant here. Thus, at sufficiently high temperatures $R_1(\omega)$ is indeed expected to follow eq 17 at low frequencies. With given spin density[27] n_s (cf. Table 1), the diffusion coefficient D can be directly extracted from $R_1(\omega)$ measured at different temperatures. Important to note, this approach yields absolute $D(T,M)$ values without resorting to a master curve construction.

2.2. Mean-Square Displacement Derived from FG NMR. A stimulated echo FG NMR experiment measures an autocorrelation function, $S_{\text{Dif}}(\tau,t)$, which is essentially equivalent to the intermediate incoherent scattering function in neutron scattering.[28] The evolution time τ is proportional to the size of the scattering vector $Q = \tau\gamma g$ (gyromagnetic ratio γ, magnetic field gradient g), and the mixing time t is the dynamic time variable. Since FG NMR always deals with the hydrodynamic regime (Q^{-1} small compared to intermolecular distances) for isotropic diffusion this correlation function is in the limit $t \gg \tau$ given by[6,29]

$$S_{\text{Dif}}(\tau, t) = \exp[-g^2\gamma^2\tau^2 D(t)t] \tag{18}$$

The time-dependent diffusion coefficient $D(t)$ accounts for cases where a possibly sublinear time dependence of the mean-square displacement is mapped onto $D(t)$ via

$$\langle r^2(t)\rangle = 6D(t)t \tag{19}$$

By measuring in two different gradients, G and g, at the same Larmor frequency, the same evolution times τ and mixing times t, it is possible to divide the stimulated echo signals, in the following equations named S_G and S_g, by each other to eliminate all other contributions to the time-dependent NMR signal:

$$S(\tau, t) \equiv \frac{S_G(\tau, t)}{S_g(\tau, t)} = \exp[-(G^2 - g^2)\gamma^2\tau^2 D(t)t] \tag{20}$$

For the case of anomalous diffusion with a Gaussian propagator (eq 8) the corresponding expression is more involved but can, for arbitrary values of τ and t, be derived from eq 29 of ref 30:

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$$S(\tau, t) = \exp\left[-(G^2 - g^2)\gamma^2 \frac{\{D(t+2\tau)(t+2\tau)^{\alpha+2} + D(t)t^{\alpha+2} - 2D(t+\tau)(t+\tau)^{\alpha+2} - 2D(\tau)\tau^{\alpha+2}\}}{(1+\alpha)(2+\alpha)}\right] \tag{21}$$

For $t \gg \tau$, the right-hand side can be expanded and approximated leading to

$$S(\tau, t) = \exp\left[-(G^2 - g^2)\gamma^2 D(t)\tau^2 t \times \left\{1 - \frac{2}{(1+\alpha)(2+\alpha)}\left(\frac{\tau}{t}\right)^{\alpha} + \alpha\left(\frac{\tau}{t}\right)...\right\}\right] \tag{22}$$

For the case of normal diffusion, $\alpha = 1$, and in the limit $t \gg \tau$ this expression is exactly equal to eq 20. If the condition $t \gg \tau$ does not hold, eq 22 will lead to an equation quite similar to eq 20 but t being replaced by $t + {}^2/_3\tau$.[31]

Note also that the initial decay of the expression (eq 21) is the same for any model of anomalous diffusion.[11] Obviously, when deriving $D(t)$ from a full experimental decay curve according to eq 21, some ambiguity prevails due to the lacking knowledge of α. This ambiguity is subsequently transferred, via eq 19, to $\langle r^2(t)\rangle$. As pointed out in the Introduction in the relevant regimes III and IV, ${}^1/_2 < \alpha < 1$ is expected. This point and its implications will be discussed in more detail further below.

3. EXPERIMENTAL SECTION

3.1. Polymers. The polymers investigated are polybutadiene (PB) with molar masses $M_w = M = 232$k and $M = 196$k and polydimethylsiloxane (PDMS) with $M = 0.9$k, $M = 0.6$k, $M = 22$k, and $M = 232$k. The entanglement molar masses are $M_e = 1.9$k for PB and $M_e = 12$k for PDMS.[32] All polymers were purchased from Polymer Standards Service (Mainz, Germany) and used without further purification. They were filled into standard 5 mm NMR tubes and kept under vacuum for 48 h to remove paramagnetic oxygen gas.

3.2. Extraction of the Intermolecular Relaxation Rate from FC NMR. The ^{1}H NMR spin-relaxation data analyzed in this paper were measured over the temperature interval of 200−400 K using two different electronic FC relaxometers: a commercial one (Stelar Spinmaster FFC 2000) at Bayreuth University covering the ^{1}H Larmor frequency range between $\nu = \omega/2\pi$ of 10 kHz and 20 MHz and a home-built one at Darmstadt University reaching further down to about 100 Hz and up to 30 MHz. For further technical details of the Darmstadt relaxometer we refer to previous publications.[14−16]

The intermolecular relaxation rate of the measured set of $R_1^{\text{inter}}(\omega)$ was determined in our previous publications in terms of a susceptibility master curve $\chi''_{\text{inter}}(\omega\tau_s)$,[10,12,19] with the susceptibility representation given by $\chi''(\omega) \equiv \omega R_1(\omega)$. In order to isolate the inter part from the total relaxation, protonated PB was successively diluted in deuterated PB which suppresses the intermolecular relaxation, since the dipolar interaction between protons and deuterons can be neglected.[9] Extrapolating to $c \to 0$ consequently yields $R_1^{\text{intra}}(\omega)$, and the intermolecular relaxation rate follows from subtraction $R_1^{\text{inter}}(\omega) = R_1^{\text{total}}(\omega) - R_1^{\text{intra}}(\omega)$. For PDMS we showed that for $\omega\tau_s \ll 1$ the total relaxation is dominated by the intermolecular relaxation.[10] Hence, the total relaxation can be directly used in eq 13. Next, as in the case of many rheological experiments[32] frequency−temperature superposition (FTS) is exploited to cover an effectively larger frequency window compared to a single experiment.[33] As the segmental relaxation (α-process) is also covered by our NMR experiments the susceptibility master curve is scaled according to $\chi''(\omega\tau_s)$. Using $\tau_s(T)$ values[33] obtained from the construction of the master curves, the rate $R_1^{\text{inter}}(\omega,T)$ for any desired temperature T can be calculated via

$$R_1^{\text{inter}}(\omega, T) = \frac{\chi''_{\text{inter}}(\omega\tau_s)}{\omega\tau_s}\tau_s(T) \tag{23}$$

Transforming the relaxation data into the msd a cosine transformation of $R_1^{\text{inter}}(\omega)$ has to be performed according to eq 13. For that purpose a Filon-type algorithm[34] was applied.

3.3. Field-Gradient Experiments. All FG measurements were performed in a (static) superconducting gradient magnet at a ^{1}H Larmor frequency of 99.5 MHz. The magnet provides two different gradients at this frequency: A large one of $G = 168(T/m)$ and a smaller one of $g = 60(T/m)$.[35] The magnet resides on an air cushion, so that building and ground vibrations are strongly damped. Thereby, motion of the sample relative to the magnet is effectively avoided which otherwise could easily be misinterpreted in term of molecular diffusion inside the sample.[36] The position of the probe head in the magnet can be adjusted by a stepper motor. In this way, measurements in the different gradients can be performed directly after each other without removing the sample from the cryostat nor changing tuning or matching of the probe. The temperature is stabilized in a cryostat, where the temperature drift is less than 0.1 K with an absolute uncertainty of about 1 K. A 2 kW amplifier was used in order to get short ($\pi/2$) pulses of 0.4 μs length. A stimulated echo pulse sequence (90°−τ−90°−t−90°−τ−) was used to measure the time-dependent diffusion coefficient $D(t)$. The stimulated echo signal $S_{\text{StE}}(\tau,t)$ with the initial amplitude S_0 may be decomposed along

$$S_{\text{StE}}(\tau, t) = S_0 S_{\text{SSR}}(\tau) S_{\text{SLR}}(t) S_{\text{Dip}}(\tau, t) S_{\text{Dif}}(\tau, t) \tag{24}$$

where $S_{\text{SSR}}(\tau)$ and $S_{\text{SLR}}(\tau,t)$ describe the spin−spin and the spin−lattice relaxation, respectively, $S_{\text{Dip}}(\tau,t)$ denotes the dipolar correlation, and $S_{\text{Dif}}(\tau,t)$ is the diffusion-induced decay.[6,22,29] The only factor depending on the magnetic field gradient is the diffusion factor $S_{\text{Dif}}(\tau,t)$. Therefore, when dividing experimental data $S_{StE}(\tau,t)$ gained at a large gradient G by those gained in a small gradient g, we are left with $S(\tau,t)$ given by eqs 21 and 22.

Figure 1 exemplarily shows data taken at 292 K of the two normalized decay curves $S_G(\tau,t = 51\text{ ms})$ and $S_g(\tau,t = 51\text{ ms})$,

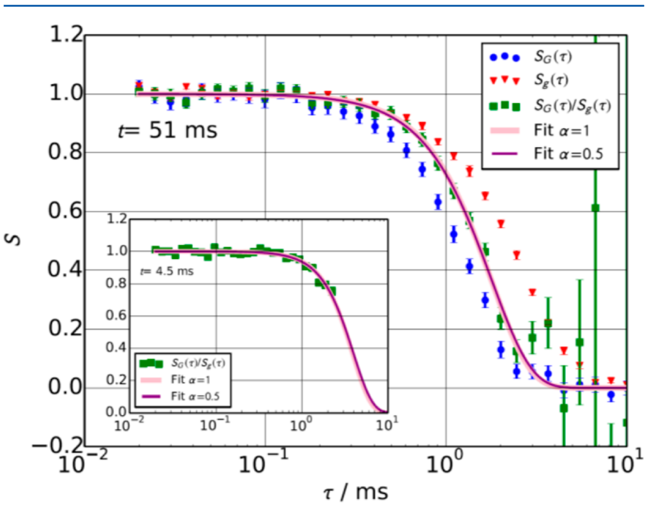

Figure 1. Stimulated echo amplitude of PDMS 232k at $T = 292$ K in two different gradients $G = 168$ T m^{-1} and $g = 60$ T m^{-1} at a ^{1}H Larmor frequency of 99.5 MHz and a fixed mixing time $t = 51$ ms. In addition, the ratio of both curves is shown. The fits of $S(\tau) = (S_G(\tau)/S_g(\tau))$with eq 22 for $\alpha = 0.5$ and $\alpha = 1$ are almost indistinguishable (drawn curves). Inset: same experiment using $t = 4.5$ ms.

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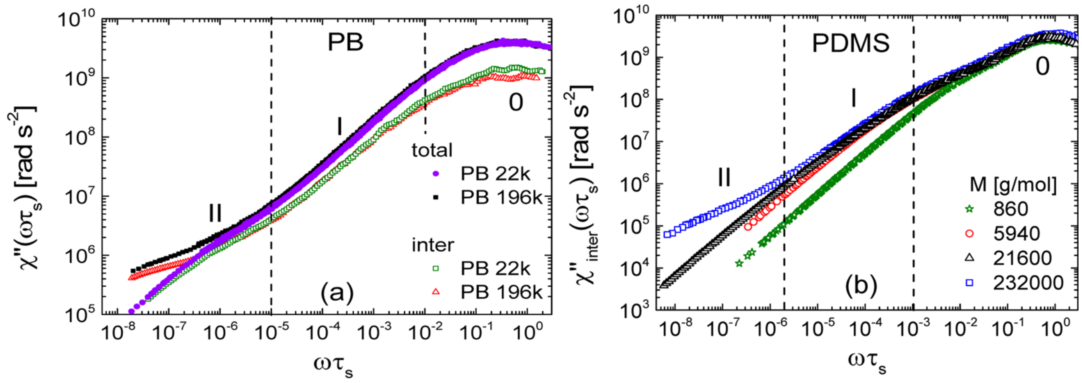

Figure 2. (a) Susceptibility master curves of polybutadiene (PB) with two different molar masses as obtained from field-cycling ^{1}H NMR relaxometry. Both the intermolecular, $\chi''_{\text{inter}}(\omega\tau_s)$, and total susceptibilities, $\chi''(\omega\tau_s)$, are shown. Different relaxation regimes (0, I, II) and molar masses (M) are indicated. (b) Susceptibility master curves $\chi'' \cong \chi''_{\text{inter}}$ of PDMS with different M.

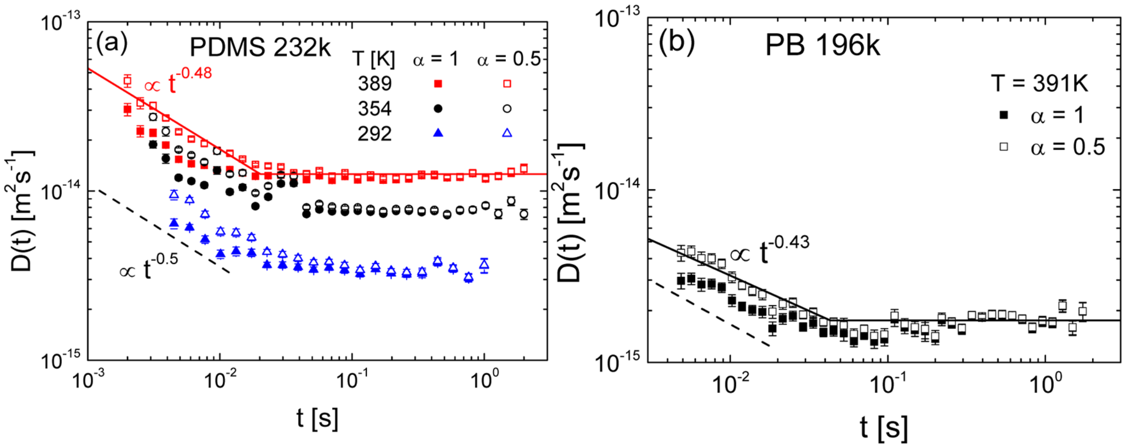

Figure 3. Time-dependent diffusion coefficients of PDMS 232k (a) and of PB 196k (b) as deduced from fits of eq 22 to the measured decay curves $S(\tau)$ as exemplified in Figure 1. For small diffusion times the resulting $D(t)$ depends on the choice of α. The solid lines show fits for highest temperatures and $\alpha = 0.5$ used in eq 22 (see text) with a power-law decay (resulting exponents are indicated) at short and a constant regime D at long times. The dashed lines illustrate a power law $D(t) \propto t^{-0.5}$ (regime III) predicted by the TR model.

measured in the two gradients G and g, respectively. Also, the ratio $S(\tau,51\text{ ms}) = (S_G(\tau,51\text{ ms})/S_g(\tau,51\text{ ms}))$ is plotted.

Note that the condition $t \gg \tau$ is fulfilled for this example as S reaches almost zero at $\tau \approx 5$ ms $\ll t = 51$ ms. If this condition was not fulfilled, the diffusion time t would not be defined accurately since the molecules significantly diffuse during the evolution time τ. In order to accounting for the condition $t \gg \tau$, we decided to evaluate the data only for evolution times $\tau < 0.5t$. In this limiting case, the "true" diffusion time may still be approximated by t as a lower limit. An upper limit for the diffusion time can be approximated by $t + \tau_{\text{max}}$. Here τ_{max} is taken either to be the longest evolution time used for the fit or the half decay time of the magnetization, depending on which one is shorter. The inset of Figure 1 shows the diffusion decay at the shortest chosen diffusion time, $t = 4.5$ ms, of PDMS 232k at $T = 292$ K. According to the above condition only the beginning of the decay curve is shown and fitted. From these fits time-dependent (apparent) diffusion coefficients $D(t)$ are gained which will be discussed in section 4.2.

4. RESULTS

4.1. Susceptibility Master Curves from FC NMR.

Figure 2 displays the susceptibility master curves of PB (a) and PDMS (b), both extending over about 9 decades in frequency, as obtained from FC ^{1}H NMR relaxometry comprising relaxation data in the temperature range of around 200−400 K (cf. ref 10). In the case of PB both the total relaxation data $\chi''(\omega\tau_s)$ and the pure intermolecular contributions $\chi''_{\text{inter}}(\omega\tau_s)$ are shown, the latter being extracted previously by isotope dilution experiments.[10,19] The susceptibility shows three relaxation regimes denoted by 0, I, and II. Regime 0 reflects the (fast) segmental dynamics and manifests itself in a relaxation peak at $\omega\tau_s \cong 1$. For $\omega\tau_s \ll 1$ regime I features a M-independent power-law regime reflecting Rouse dynamics while regime II at lowest frequencies ($\omega\tau_s \ll 10^{-5}$ in PB and $\omega\tau_s \ll 2 \times 10^{-6}$ in PDMS) is characteristic for entanglement dynamics and has a significantly smaller and M-dependent power-law exponent.[10,19,37] We note that the crossover from regime 0 to I, i.e. the beginning of the Rouse regime, is somewhat difficult to define. It works better for the transition between I and II occurring at $\omega\tau_e \approx 1$ (see below). As far as PDMS is concerned, it was shown that the relaxation is dominated by intermolecular relaxation over the whole frequency range $\omega\tau_s \ll 1$,[10] implying $\chi'' \cong \chi''_{\text{inter}}$. Another peculiarity is that for PDMS the power-law regime I first appears at frequencies ($\omega\tau_s < 10^{-3}$) of 1 decade smaller than for PB ($\omega\tau_s < 10^{-2}$). Actually, some kind of shoulder is recognized. We will return to this point below.

4.2. Time-Dependent Diffusion Coefficients Gained from FG NMR.

Let us proceed with the results from FG NMR. For all temperatures the diffusion coefficients depicted in Figure 3 for both PDMS 232k and PB 196k clearly show a time dependence at short times. One subtlety has to be taken care of while fits with eq 22 lead to decay curves which cannot

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Table 1. Collected Properties of PB and PDMS at 390 K[a]

property	PDMS 232k (T = 389 K)		PB 196k (T = 391 K)	
τ_1 [ms]	20		43	
D [m^2 s^{-1}]	1.3×10^{-14}		1.8×10^{-15}	
$\langle r^2(\tau_1)\rangle$ [nm^2]	1500		450	
$\langle R_0^2\rangle$ [nm^2]	1100[40]		1700[40]	
	experimental (present work)	from literature	experimental (present work)	from literature
a_0 [nm]	4.8[41]	7.9,[40] 7.2[42]	3.2	3.2,[43] 4.4[40]
b [nm]	1.6	1.3[32]	0.5	1.0[32]
M_0 = [g mol^{-1}]	381[32]		113[32]	
M_e = [g mol^{-1}]	12k[32]		1.9k[32]	
$N_e = M/M_e$	31		17	
$N = M/M_0$	609		1735	
T_g [K]	150[44]		180,[45] 177,[19] 175[46]	
n_s [10^{28} m^{-3}]	4.70[27]		5.75[27]	

[a]Terminal relaxation time τ_1 and $D = D(t\to\infty)$ determined from Figure 3; the msd $\langle r^2(\tau_1)\rangle$ and the mean-square end-to-end distance $\langle R_0^2\rangle$ taken from ref 40. Estimates for the tube diameter a_0 and the Kuhn length b and in comparison to values given in the literature (as indicated); for their uncertainties see text. Other parameters used were taken from the literature: the Kuhn molar mass M_0, the entanglement molar mass M_e, the ratios $N_e = M/M_e$ and $N = M/M_0$, the glass transition temperature T_g, and the spin density n_s.

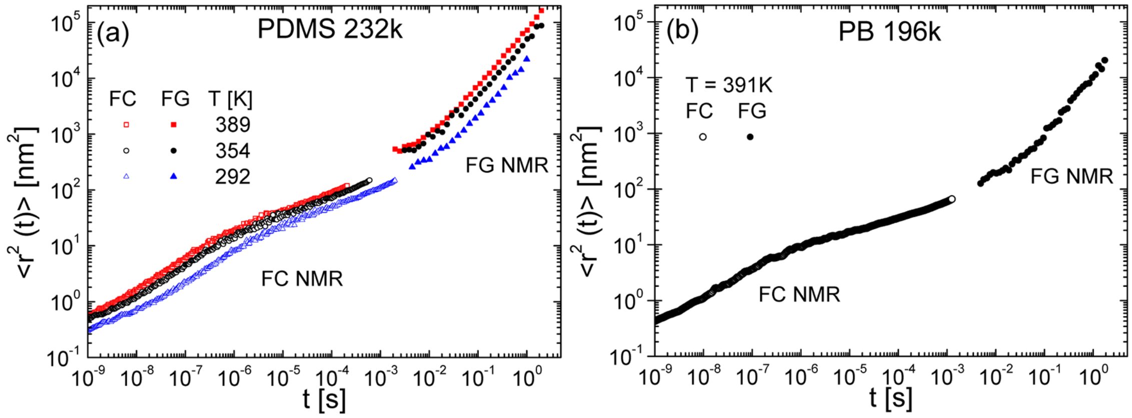

Figure 4. (a) Segmental mean-square displacements obtained from field-cycling (FC) and field-gradient (FG) ^{1}H NMR for PDMS 232k at temperatures as indicated. (b) Corresponding results for PB 196k.

distinguish between α = 0.5 (TR regime III) and α = 1 (TR regime IV); for small diffusion times the resulting apparent diffusion coefficients $D(t)$ do differ and for α = 0.5 being above those for α = 1 by up to a factor of 2 (Figure 3).

The short-time regime in both polymers can be acceptably described by a power law $D(t) \propto t^{-0.5\pm0.1}$ which is expected for regime III of the TR model (cf. the inset of Figure 5b). At long times the diffusion coefficient becomes time-independent when the terminal relaxation of free diffusion (regime IV) is finally reached. For highest temperatures we exemplarily included fit curves for both polymers in Figure 3 consisting of an initial power-law decay regime and a constant long time regime $D(t\to\infty) = D$ the transition occurring at the terminal relaxation time τ_1. The resulting D and τ_1 are included in Table 1. Although the transition between regimes III and IV is clearly discernible, the experimental errors are significant. We emphasize again the experimental difficulties of reaching such low diffusion coefficients by FG NMR, especially in the case of PB (cf. Experimental Section). At this point a comment on the validity range of FG measuring time-dependent diffusion coefficients is appropriate: A competing process which may eventually feign segmental diffusion or at least contribute to $D(t)$ is spin diffusion. The validity criterion chosen by us requires the long time diffusion coefficient to scale with the inverse segmental relaxation time, i.e., $D \propto \tau_s^{-1}$. If we do so, it turns out that data only at those temperatures shown in Figure 3 can be trusted; at lower temperatures spin diffusion gradually takes over.

As stated, in regime IV the evaluated $D(t)$ does not depend on the choice of the parameter α in eq 22, but it does in regime III. Here, α = 0.5 is expected from the TR model for which we will base the evaluation of the msd on those $D(t)$ data obtained with α = 0.5. According to eq 19 the time-dependent msd, to be discussed later, can now be directly evaluated from these $D(t)$.

4.3. Mean-Square Displacements from Combination of FC and FG Data. In Figure 4 the msd of both PDMS 232k (a) and PB 196k (b) is displayed for several temperatures as obtained from the application of eq 13 (FC NMR data) and via $\langle r^2(t)\rangle = 6D(t)t$ (FG NMR data). For the FC NMR data we employed α = 3/8 being the average of α = 0.5 and α = 0.25 predicted for the regimes I and II of the TR theory, respectively, which are usually covered by the FC method for high M. We once again emphasize that the influence of the parameter α is marginal here. About 10 decades in diffusion time is covered, and the data obtained from FC NMR agree well with the absolute values derived from FG NMR. Several power-law regimes are identified in both polymers and will be discussed below. We note that at shortest times, say, at $t < 10^{-9}$ s, eq 13 does no longer apply as the spatial extension of the monomer unit is not taken into account. Thus, our analysis holds for times $t \gg \tau_s$.

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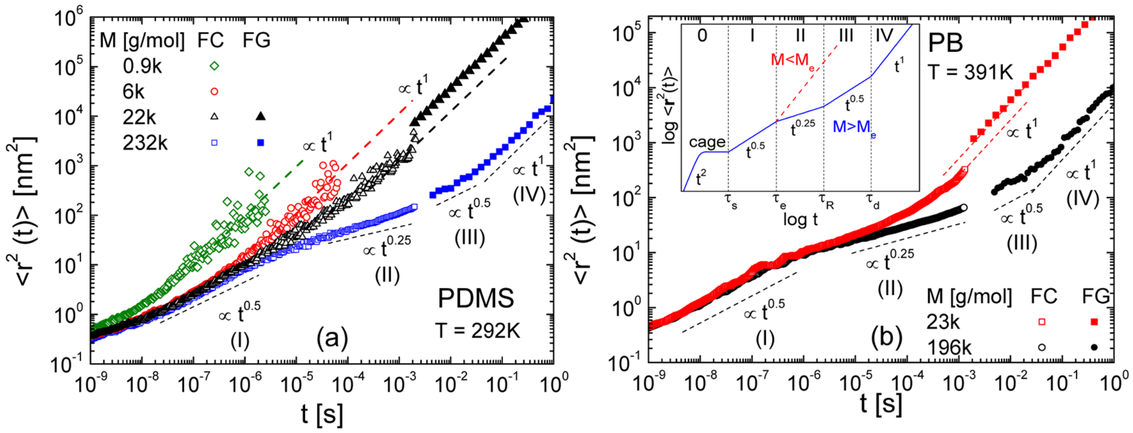

Figure 5. Segmental mean-square displacement of PDMS (a) and PB (b) obtained from FC and FG NMR; data for a single temperature are shown. In both, data for short-chain systems are included. Dashed lines represent corresponding power-law regimes. The inset in (b) illustrates the prediction of the TR theory.

Figure 5 shows the msd of PDMS at 292 K (a) and PB (b) at 391 K and for different molar masses. First, we discuss the case of highest M, PDMS 232k and PB 196k, both being well above M_e. Starting with the FC NMR data at short times, the msd of both polymers follows a power-law which matches the prediction of (free) Rouse dynamics with $\alpha_R = 0.5 \pm 0.1$ well (regime I). At succeeding times, in regime II a power-law exponent of $\alpha_{II} = 0.3 \pm 0.1$ is observed in PDMS 232k while $\alpha_{II} = 0.2 \pm 0.1$ is found in PB 196k. Both values are consistent with 0.25 of the constraint Rouse regime (II) predicted by the TR model. On even longer times, a further power-law with $\alpha_{III} = 0.5 \pm 0.1$ for both PDMS 232k and PB 196k appears in the FG NMR data. This relaxation regime is quite limited in time and actually difficult to be unambiguously identified. However, it can be well estimated from the diffusion data shown in Figure 3. Finally, at longest times the free diffusion limit (regime IV) with an exponent $\alpha_{IV} = 1.0 \pm 0.1$ is reached in both polymers. Thus, for the first time all four power-law regimes of the TR model are discovered in two high-M polymers by combining two NMR techniques, and the power-laws are close to those predicted by the TR model.

When going to lower M the regimes II and III, which reflect the entanglement dynamics, shrink and the free diffusion regime (IV) directly succeeds the Rouse regime (I). PDMS 22k and PDMS 6k, for instance, are close to $M_e = 12$k. While the Rouse regime (I) is still fully established with $\alpha_R = 0.5 \pm 0.1$ in the FC NMR data, regime II disappears in both cases, and already the FC NMR measurements reach the regime of terminal relaxation, i.e., $\langle r^2(t \gg \tau_1)\rangle \propto t^1$. Regime III is completely absent in the FG NMR data of PDMS 22k. Although FC NMR and FG NMR data show virtually the same exponent in the terminal regime (IV), they do not fully agree on an absolute scale provided by the FG data; a discrepancy of a factor 3−4 prevails. We will discuss possible reasons for this mismatch below. For PDMS 0.9k even the Rouse regime is missing. In this case essentially no polymer dynamics is discovered as this oligomer actually behaves like a simple liquid. For PB just one further molar mass with $M = 23$k is studied by FC NMR as an isotope dilution experiment is necessary here in order to isolate the intramolecular relaxation.[9,10,19−21] Although regime II is significantly less pronounced, it can still be identified. Regime III appears to be absent in the corresponding FG NMR data. While for $M \gg M_e$ only the combination of FG and FC NMR provides the full msd, FC NMR already suffices to probe the translational dynamics for $M \leq M_e$.

In Figure 6, we compare the msd for PB 196k and PDMS 232k at virtually the same temperature ($T \cong 390$ K) in a common presentation. From the intersection points of the power-law regimes I and II we extract rough estimates of the tube diameter[1] $a_0 = \langle r^2(t{=}\tau_e)\rangle^{1/2}$ and the entanglement time τ_e. We read from the plot $a_0 = 4.8$ nm for PDMS and $a_0 = 3.2$ nm for PB (cf. also Table 1). A comparison with literature data follows in the Discussion. Moreover, the Kuhn length b can be estimated via[1] $\langle r^2(\tau_1)\rangle = Nb^2$, i.e., from the intersection points between regimes III and IV. In order to get b the number of Kuhn segments $N = M/M_0$ with M_0 being the Kuhn segment mass has to be known. We tentatively adopt M_0 from the literature[32] (cf. Table 1). The extracted proxies are $b = 1.6$ nm for PDMS and $b = 0.5$ nm for PB. The uncertainties of both a_0^2 and b^2, as read from Figure 6, are estimated to be of about a factor of 2.

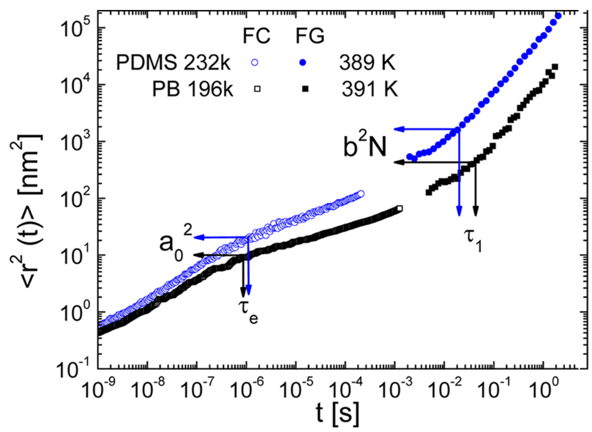

Figure 6. Segmental mean-square displacement (msd) of polydimethylsiloxane (PDMS) and polybutadiene (PB) for a similar temperature. Entanglement time τ_e and disengagement time $\tau_d = \tau_1$ referred to the intersection points of the power-law transitions between regimes I to II and III to IV, respectively, are indicated. The msd evaluated at τ_e and τ_1 provide an estimate of the tube diameter a_0 and segment length b, respectively (cf. Table 1).

Recently,[27] in the case of PDMS absolute (long-time) diffusivities $D(M)$ were determined by applying eq 17; i.e., the

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total relaxation rate was analyzed in the low-frequency limit for various temperatures without applying FTS. In Figure 7 we compare those data with the ones from the present FG NMR work and with literature data[6] at about 292 K. All three data sets match very well. While FG NMR covers the entanglement regime $M > M_e$ and a power-law with the exponent $\alpha = 2.3 \pm 0.2$ is found in agreement with literature data,[5,6,38,39] FC NMR cannot cover high M with the present low-frequency limit of some 100 Hz. In any case, the crossover from Rouse to entanglement dynamics can be well identified. We emphasize that the low-M D values are not corrected for the M-dependent monomeric friction coefficient as it was done before.[27] In the case of PB we measured only at a single M value and literature data at this temperature is not available.

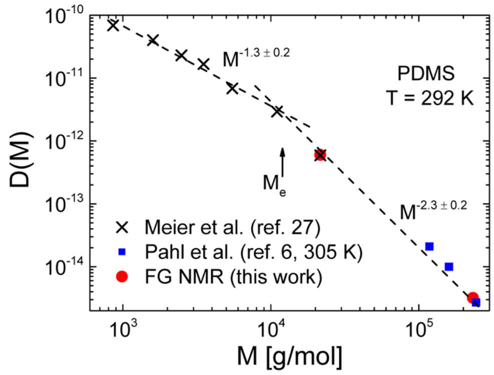

Figure 7. Molar mass dependence of the (long-time) diffusion coefficient D of PDMS at 292 K obtained from FG NMR (red disks). For comparison, values of Meier et al.,[27] derived from FC ^{1}H NMR relaxometry (black crosses) and from Pahl et al.,[6] also obtained by FG NMR (blue squares), yet, at a slightly higher temperature of 305 K, are included. Two power-law regimes are indicated while the crossover molecular mass agrees well with the entanglement molar mass $M_e \approx$ 12k (cf. Table 1 and refs 32 and 40).

5. DISCUSSION

The complementary frequency/time windows of FC ^{1}H NMR and static FG ^{1}H NMR experiments were combined in order to measure the segmental msd in two different polymers, namely PDMS and PB, over almost 10 decades. While the msd can be easily obtained from the (time dependent) diffusion coefficient measured by FG NMR, it is calculated from FC ^{1}H NMR intermolecular relaxation data involving Fourier transformation (eq 13). In general, the intermolecular contribution must be isolated from the total ^{1}H spin−lattice relaxation data by applying an isotope dilution experiment.[9,10,19−21] Only in favorable cases like PDMS the total ^{1}H relaxation at $\omega\tau_s \ll 1$ is dominated by the intermolecular relaxation since the fast methyl group rotation reduces the intramolecular relaxation. Thus, the data can directly be used for obtaining the msd.[10] As the time windows of FC and FG NMR do not overlap, the experiments need to be performed at different temperatures; thus, a direct comparison of the msd relies in the construction of the master curve $\chi''_{\text{inter}}(\omega\tau_s)$ which assumes the validity of FTS. With $\tau_s(T)$ known from the master curve construction, $\chi''_{\text{inter}}(\omega)$ can be calculated for any temperature. As will be discussed below, there are strong indications that FTS indeed applies well for both polymers.

At time scales longer than the segmental relaxation time ($t \gg \tau_s$) four different relaxation regimes, each showing power-law behavior, can be identified for the msd in the high-M limit and compared to the TR model.[1] While regimes I and II are covered by FC NMR, FG NMR traces the transition from regime III to IV (Figure 5). The four regimes can be associated with those from the TR model, namely that of the (free) Rouse (I), the constrained Rouse (II), the reptation (III), and finally the free diffusion (IV) regime at longest times. All found power-law exponents agree well, within ±0.1, with the predicted ones. On an absolute scale, it appears that there is some mismatch between the msd measured by FC and FG NMR, in particular for M = 22k for both PDMS as well as PB (cf. Figure 5). Field-gradient NMR yields absolute msd values within an uncertainty of a factor of 2 at most which is due to the lacking knowledge of the power law exponent α. The corresponding msd data of FC ^{1}H NMR relies on several assumptions: (i) Adopting $\tau_s(T)$ from master curves assumes FTS to hold. Any correction will shift the FC ^{1}H NMR msd along the time axis (cf. Figure 5). (ii) The choice of the numerical factor δ in eq 13 weakly influences the value of the msd. It is actually time dependent due to the different diffusion regimes covered. (iii) The assumed relation between relative and absolute msd may not strictly hold (cf. Theoretical Background). In general, it may vary according to $\langle r^2(t)\rangle = (1/q)\langle \tilde{r}^2(\tau_1)\rangle$ with $0 \le q \le 2$, the limits representing the cases of rigidly bonded segments and completely uncorrelated segmental motion, respectively. However, when regime IV is reached, q = 2, precisely. (iv) The Gaussian approximation is assumed for deriving the connection between the msd and $R_1(\omega)$ (eq 8). If it does not fully apply an additional numerical factor would appear. (v) Fast dynamics (β-process, methyl group rotation in PDMS, etc.) also may slightly affect the results. Given these various sources of possible corrections of the FC ^{1}H NMR data, we think the present results provide a fair agreement between the two NMR methods. So, within these limitations, for the first time, all four diffusion regimes of the msd are covered and confirm the TR model. We note that the power-law regimes I and II were also covered by NS experiments.[4] Here, FC NMR and NS are competing techniques, yet according to our knowledge no msd data from NS addressing the regime I to II transition are available for PDMS and PB.

The TR model also provides predictions[47] regarding the segmental reorientation correlation function $A_l(t)$ of rank l. According to the so-called return-to-origin hypothesis,[47,48] the model predicts an inverse proportionality between $A_l(t)$ and $\langle r^2(t)\rangle$ in regimes II and III independent of rank l, specifically, $A_l(t) \propto \langle r^2(t)\rangle^{-1}$. This relation was tested by FC ^{1}H and ^{2}H NMR and found to be violated.[10] However, in a recent publication by Furtado et al. applying also isotope dilution and double quantum (DQ) NMR together with MD simulations the TR predictions were confirmed essentially.[37] Yet, in a succeeding publication[19] we reanalyzed our results together with new measurements. We critically assessed the findings and once again reproduced our previous results. For example, in the case of PB the exponent of $A_2(t)$ in regime II turns out to be $\alpha_{\text{II}} \approx 0.44$, which is in strong disagreement with the prediction 0.25 given by the TR model, although finite chain length effects were taken into account. We emphasize that $A_2(t)$ can only be obtained from isolating the intramolecular ^{1}H relaxation contribution or, more directly, by FC ^{2}H NMR. Although the discrepancy among the results from DQ and FC NMR still needs to be settled, given our FC NMR results, it appears that the inverse proportionality between $A_2(t)$ (probing reorientation) and the msd (probing translation) does not apply.

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Possibly, the relation depends on the microscopic details of the polymer chain, i.e., the chemical structure of the monomer.

At 390 K the tube diameter is obtained from the condition $a_0 = \langle r^2(\tau_e)\rangle^{1/2}$, and τ_e is defined as the crossover time between the regimes I and II. In the case of PDMS the deduced diameter, a_0 = 4.8 nm, is smaller than reported from rheological experiments which gave a_0 = 7.9 nm[40] while a_0 = 7.2 nm was found in NS experiments (cf. Table 1 and Figure 6).[42] We note that in the case of PDMS our value differs somewhat from the one reported previously.[10,41] Regarding PB, our value of a_0 = 3.2 nm is in good accordance with reference values derived from rheological experiments (cf. Table 1).[40,43] It appears, however, that a_0 slightly depends on temperature.[40,49]

The FG NMR data also allow estimating the Kuhn lengths b from the transition time between regime III and IV using $\langle r^2(\tau_1)\rangle = (M/M_0)b^2$.[1] The value b = 1.6 nm obtained for PDMS agrees reasonably well with literature data (cf. Table 1). In PB, for which b = 0.5 nm is found, there is a deviation by a factor of 2. Of course, M_0 and b are not independent of each other, and therefore only a proxy for b can be given. On the other hand, within the TR model,[1,50] at τ_1 the msd is expected to be equal to the mean-square end-to-end distance $\langle R_0{}^2\rangle$, i.e., $\langle R_0{}^2\rangle \approx \langle R^2(\tau_1)\rangle$. It can be determined by small-angle neutron scattering (SANS) and compared to our results (cf. Table 1). Given the molar masses of PB and PDMS investigated in this work the SANS results reported in ref 40 provide $\langle R_0{}^2\rangle$ = 1100 nm^2 for PDMS and $\langle R_0{}^2\rangle$ = 1700 nm^2 for PB, while we obtain $\langle R^2(\tau_1)\rangle$ = 1500 nm^2 and $\langle R^2(\tau_1)\rangle$ = 450 nm^2. Again, the discrepancy is much less for PDMS than for PB. Here one has to keep in mind that data from experiments characterizing the dynamics (FG NMR) and the structure (SANS) may not necessarily agree.

Next, we compare the time constants $\tau_s(T)$, $\tau_e(T)$, and $\tau_1(T)$ from our work with data found in the literature. As far as PB is concerned, we first display the segmental time constant $\tau_s(T)$ in Figure 8 as obtained from combining FC ^{1}H NMR and dielectric spectroscopy[33,51] (DS) for various high-M samples. In this limit $\tau_s(T)$ is independent of M which goes along with a saturation of T_g. Usually in polymers, $\tau_s(T)$ is interpolated by a Vogel–Fulcher–Tammann equation. Alternatively, we use a recently introduced four-parameter function[52,53] (cf. Appendix). Experimental data of high-temperature segmental time constants are scarce. Here, FC NMR offers a unique opportunity to provide $\tau_s(T)$. The values $\tau_s(T)$ can be checked against those from rheological experiments which usually provide the temperature dependence in terms of the shift factor a_T. In ref 46, $a_T(T)$ is given for PB (M = 130k), and together, with τ_s(298 K) = 0.3 ns,[32] it is referred to $\tau_s(T)$ as described in the Appendix. The thus calculated curve (green dashed line) reproduces our data almost perfectly. Alternatively, $\tau_s(T)$ (more precisely the fastest Rouse time $\tau_0(T)$) can be calculated from the monomeric friction coefficient $\xi(T)$ given in ref 45 (cf. eq 27, Appendix). This leads again to a fair agreement with our results (cf. purple dashed line in Figure 8).

Rheological experiments are well established for determining the terminal time τ_1. We follow ref 32 and estimate τ_1 from the intersection of the storage and loss shear moduli at lowest frequencies. The red (open and full) symbols in Figure 8 represent τ_1 values obtained in this way from a multitude of rheological results.[32,46,54−57] A large range of molar masses 21k $\leq M \leq$ 925k is covered, essentially at room temperature. Assuming FTS, our interpolation (black solid line) of the FC NMR and DS data along eq 25 for $\tau_s(T)$ is shifted to intersect

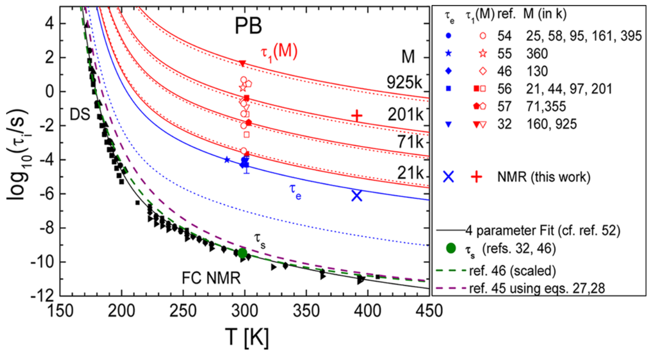

Figure 8. Temperature and M dependence of the time constants of PB: $\tau_s(T)$ for several high-M samples ($M > M_e$) (distinguished from each other by the different black symbols) as obtained from FC NMR and DS,[33,51] respectively, interpolated by the black solid line in comparison to rheological data reported in the literature as indicated (green and purple dashed lines). Terminal relaxation time $\tau_1(T)$ as obtained by FG NMR (red cross) and by rheology (other red symbols). Solid red lines: same interpolation of $\tau_s(T)$ but shifted for hitting selected M values (represented by solid red symbols) suggesting the behavior of $\tau_1(T)$. Red dashed lines: prediction for $\tau_d(T)$ by the TR model for the selected M. τ_e(391 K) as obtained by FC NMR (blue cross) in comparison to rheological data (other blue symbols). The solid blue line is another shifted version of the interpolation of $\tau_s(T)$. Dashed blue line: prediction by the TR model for $\tau_e^{\mathrm{DE}}(T)$. Molar masses M are indicated.

selected data points (solid red symbols in Figure 8) obtained from rheology providing the solid red lines. Obviously, our FG NMR measurement, yielding τ_1(391 K) = 43 ms (cf. Figure 3b) in the case of PB 196k, is in fair agreement with the behavior suggested by shifting $\tau_s(T)$ through the value given by the rheological data given in ref 56 for the similar molar mass of 201k. The M dependence $\tau_1(M)$ is further discussed below.

Field-cycling NMR also provides τ_e (cf. Figure 6); actually, this crossover from regime I to II is well documented in all FC NMR experiments.[10,19,33] We are not aware of any work which attempts to estimate τ_e in a model-independent way. In rheological spectra and according to the ideal TR model, τ_e should be given by the beginning of the rubber plateau in $G'(\omega)$ or $G(t)$ or likewise from the end of the Rouse power-law regime. This implies that τ_e is also given by the minimum in $G''(\omega)$ (cf. ref 50). However, experimentally the position of the minimum depends weakly on M, which appears not to be the case for τ_e as probed by FC NMR. Notwithstanding, we estimated τ_e from the minimum in $G''(\omega)$ and included it in Figure 8 (blue symbols) together with an error bar indicating the spread with M. A version of the interpolation of $\tau_s(T)$ is once again shifted to intersect (the average of) those τ_e data. Our value of τ_e(391 K) = 0.8 μs (cf. Figure 3a) is in good agreement with these rheological results. Thus, it appears that the time constants τ_e obtained from NMR agree rather well with those obtained from rheology when taking τ_e from the minimum, ignoring its weak M dependence. As will be shown below this also holds for PDMS.

The results collected for PB and shown in Figure 8 can be checked against the predictions of the TR model according to which the disengagement time τ_d is related to τ_s via[1,11] $\tau_d(T) = 3\tau_s(T)(N^3/N_e)$. Such calculated curves for selected molar masses (red solid symbols) using the interpolation of $\tau_s(T)$ and N_e given in Table 1 are included in Figure 8 as red dashed lines. Fair agreement with the experimental results (τ_1) is found.

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However, experimentally $\tau_1(M)$ rather scales with $M^{3.4}$.[32,44,46] Therefore, deviations between τ_1 and τ_d are anticipated. With respect to the entanglement time τ_e the TR model provides[1,11] $\tau_e^{DE} = \tau_s N_e^2$.This prediction is also included in Figure 8 as a blue dashed line. Likewise, it can be expressed and estimated in terms of the disengagement time along $\tau_e^{DE} = (\tau_d/3)(N_e/N)^3$. A very similar result (and thus not shown) is found to that obtained using $\tau_e^{DE} = \tau_s N_e^2$. Apparently, in the case of PB the TR prediction for τ_e^{DE} is more than 2 decades shorter than suggested by FC NMR and by the minimum of the rheological spectra, even if the more realistic $\tau_e^{DE} \propto \tau_1(N_e/N^{-3.4})$ (instead of -3.0) scaling is used. As indicated in ref 32 τ_e, when calculated from the TR model, appears to be close to the crossing of $G'(\omega)$ and $G''(\omega)$ at high frequencies which indeed is much higher than the minimum in $G''(\omega)$. Yet, it is not obvious what the physical meaning of this intersection is.

The comparison of the different time constants $\tau_s(T)$, $\tau_e(T)$, and $\tau_1(T,M)$ can be carried out for PDMS as well, which is done in Figure 9. When compared to rheological data[44] (cf.

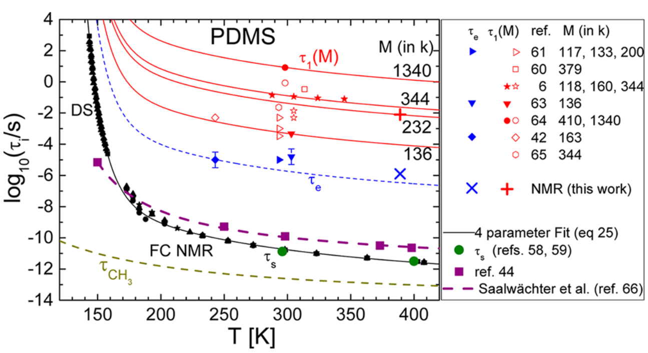

Figure 9. Temperature and M dependence of the time constants in PDMS: $\tau_s(T)$ for several high-M samples ($M > M_e$) (distinguished by the black solid symbols) as obtained from FC NMR and DS fitted with a four-parameter function[52] (black solid line, cf. Appendix) in comparison to rheological data (see text). $\tau_e(389\text{ K})$ as obtained by FC NMR (blue cross) in comparison to rheological data (other blue symbols). Dashed blue line: prediction by the TR model for τ_e^{DE}. Red: $\tau_1(T)$ as obtained from our FG NMR results (red cross) and from reference data (also FG NMR[6] and rheology, other red symbols). Solid red lines: several versions of the interpolation of our $\tau_s(T)$ data shifted for intersecting selected M values (solid red symbols) suggesting the behavior of $\tau_1(T)$. The molar masses studied in each reference are indicated. Also indicated: time constant $\tau_{CH_3}(T)$ of methyl group rotation.[60]

Appendix) our high-temperature values for $\tau_s(T)$ appear to be faster by 1 decade. Two values for $\tau_s(T)$ were reported,[58] which were originally deduced from NS[59] confirming our result (green disks in Figure 9). As a note of caution: we were unable to retrace these values to the original work.[59] As the $\tau_s(T)$ values determined by FC ^{1}H NMR could be spoiled by the influence of the methyl group rotation, we included in Figure 9 the corresponding values $\tau_{CH_3}(T)$; they are significantly shorter.[60] As another possibility we suggest the following: FC NMR and DS actually probe reorientations on molecular level (so-called α-process). This must not necessarily be identical with the shortest Rouse time τ_0 reflecting reorientation of the Kuhn element, a finding deserving further investigations. As in the case of PB $\tau_s \approx \tau_0$ is found from the consistency between eq 28 (Appendix) and the experiments, we conclude that rather $\tau_0 \approx 10\tau_s$ holds for PDMS. Reinspecting Figure 2b the power-law regime I (characteristic of free Rouse dynamics) sets in at $\omega\tau_s < 10^{-3}$ while in the case of PB this crossover is already observed at $\omega\tau_s < 10^{-2}$ (Figure 2a).

The terminal relaxation time $\tau_1(T = 389\text{ K}) = 20$ ms obtained by FG NMR for PDMS 232k essentially agrees with reference data: the value is somewhat lower than anticipated from the FG NMR data in ref 6 for PDMS 344k and somewhat larger than that of PDMS 200k measured by rheology.[61] The value $\tau_e(389\text{ K}) = 1.3\ \mu$s obtained by FC NMR is in fair agreement with data obtained from rheological experiments[42,61,63] (blue symbols in Figure 9) extrapolated to 389 K when again taking the minimum in $G''(\omega)$. As we lack of a reliable source of τ_0 we calculate τ_e^{DE} from τ_1 applying $\tau_e^{DE} = (\tau_1/3)(N_e/N)^3$ for PDMS 232k. Unlike in PB our value τ_e agrees acceptably with τ_e^{DE} predicted by the TR theory. Otherwise, the estimation along $\tau_e^{DE} = \tau_s N_e^2$ provides, as in the case of PB, values, which are by more than 2 orders of magnitude too short when compared to the experimental τ_e, even when $\tau_0 \approx 10\tau_s$ is taken into account.

Finally, the M-dependence of τ_1 evaluated at around 300 K is plotted in Figure 10. In the case of PB we included a fit with a

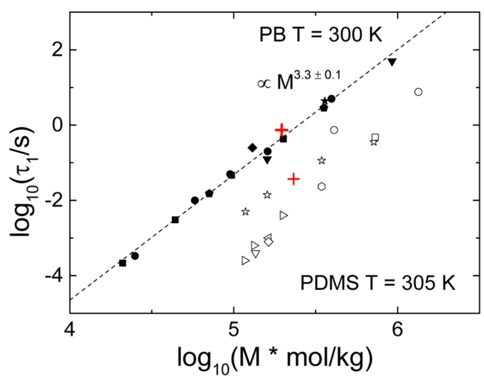

Figure 10. M-dependence of the terminal relaxation time τ_1 obtained from rheological experiments, evaluated at 300 K for PB (solid symbols) and PDMS at 305 K (open symbols). Red crosses: τ_1 as obtained from the transition between regime III and IV in Figure 6. The symbol style reflects the various references given in Figures 8 and 9, respectively. A fit with a power-law provides $\tau_1 \propto M^{3.3\pm0.1}$ for PB (dashed black line).

power-law providing $\tau_1 \propto M^{3.3\pm0.1}$ which, in accordance with other authors[44,46] and our FG NMR data point, fits in perfectly. Since for PDMS the data scatter more we omit a fit. Our data point, however, fits well to FG NMR results reported elsewhere.[6]

■ CONCLUSIONS

By combining field-cycling (FC) and field-gradient (FG) NMR experiments, the segmental mean-square displacement (msd) is determined covering 10 decades of time for PB and PDMS. While the msd can directly be obtained by FG NMR, it is calculated from the dispersion of the FC ^{1}H NMR intermolecular relaxation rate $R_1^{\text{inter}}(\omega)$ involving Fourier transformation. The quantity $R_1^{\text{inter}}(\omega)$ was isolated from the total ^{1}H spin−lattice relaxation rate by applying isotope dilution experiments. As the FG and FC NMR experiments, when performed at the same temperature, cover completely different time/frequency ranges, FTS has to be applied to convert the FC data to temperatures covered by the FG experiments. For the first time all four power-law regimes of the

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tube-reptation (TR) model are identified in the msd with the exponents rather close to the predicted ones. Even on absolute scale the msd extracted from FC NMR agrees fairly well with that from FG NMR. The estimates of the Kuhn length b and the tube diameter a_0 resemble those found in the literature. Comparing the temperature dependence of segmental time $\tau_s(T)$ and disengagement time $\tau_d(T)$ to literature data, we find satisfying agreement. In the case of PDMS we find indications that the segmental time is not identical with the shortest Rouse time; i.e., the Rouse regime I sets in about 1 decade later in time as compared to PB. In the case of τ_e, agreement with rheological data is achieved when the time constant is extracted from the minimum in the shear modulus $G''(\omega)$. Concerning the TR predictions the molar mass (M) dependence of τ_d is essentially reproduced. Yet, calculating τ_e from τ_d for PDMS yields agreement with experimental data while this is not the case for PB, for which the prediction yields times by 2 orders of magnitude too short. In no case τ_e is correctly reproduced from $\tau_s(T)$.

Altogether, we have demonstrated that with advanced equipment FC ^{1}H NMR relaxometry, which can probe the frequency dependence of the spin–lattice relaxation in a frequency range of 100 Hz–40 MHz, has become a powerful tool not only to unravel segmental reorientation as previously shown but also translational diffusion.

APPENDIX

An Interpolation Formula Used To Describe $\tau_s(T)$

As shown earlier,[52,53] the temperature dependence of the segmental correlation time $\tau_s(T)$ can be decomposed into an Arrhenius-like high-T part and in a cooperative part along

$$\tau_s(T) = \tau_\infty \exp[T^{-1}(E_\infty + E_{\text{coop}}(T))] \tag{25}$$

with

$$E_{\text{coop}}(T) = E_\infty \exp\left[-\frac{\mu}{E_\infty}(T - T_A)\right]$$

The parameters denote the high-T activation energy E_∞, an attempt time τ_∞, a generalized fragility parameter μ, and a characteristic temperature T_A. In contrast to the VFT equation it also covers temperatures well above T_g.

Extracting Segmental Time Constants from Rheological Data

In the context of discussing Figure 8, we compare $\tau_s(T)$ obtained by FC NMR and DS with rheological data. Assuming FTS,[46] shear modulus measurements on PB (M = 130k) span a broad (reduced) frequency range. This allows mapping the shift factors a_T given in ref 46 in form of a Williams–Landel–Ferry (WLF) equation

$$\log a_T = -\frac{C_1(T - T_0)}{C_2 + T - T_0} \tag{26}$$

with C_1 = 3.48 and C_2 = 163 K, to absolute values of $\tau_s(T)$: From ref 46 we estimate τ_s(298 K) = 0.3 ns corresponding to $a_{298\text{ K}} = 1$. This value is also in accordance with that given in ref 32 (green disk in Figure 8). Finally, $\tau_s(T)$ is calculated from eq 26 according to $\log \tau_s(T) = \log \tau_s(298\text{ K}) + \log a_T$.

In ref 45, the monomeric friction coefficient $\xi(T)$ of high-M PB is also given as the expression

$$\log \xi(T) = \log \xi_0 + \frac{C_1^g C_2^g}{T - T_g + C_2^g} \tag{27}$$

with $\xi_0(T) = 10^{-13.8}$ N s m^{-1},C_1^g = 13.9, and C_2^g = 44 K. Inserting $\xi(T)$ into the definition of the segmental correlation time[1]

$$\tau_0(T) = \frac{b^2\xi(T)}{3\pi^2 k_B T} \tag{28}$$

yields the purple dashed line in Figure 8. Concerning PDMS (Figure 10) $\xi(T)$ is given in ref 44 for several temperatures. Inserting these values into eq 28 yields the purple squares in Figure 9. We note that the thereby obtained $\tau_0(T)$ values for PDMS are consistent with those used by Chávez et al.[66] but differ from those obtained by FC NMR and DS (cf. Discussion).

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Notes

The authors declare no competing financial interest.

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